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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 3.3

Cramer's Rule and Volume

Cramer's Rule

Let A be an invertible $n \times n$ matrix.
For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}$$

where $A_i(\mathbf{b})$ is the matrix obtained by replacing column i of A with $\mathbf{b}$.

Example 1

Prove Cramer's Rule.

Example 2

Use Cramer's Rule to solve the system.

$$\begin{aligned}2x_1 + 6x_2 &= -1 \\ x_1 + 8x_2 &= 2\end{aligned}$$

Example 3

Determine the values of the parameter s for which the system has a unique solution, and describe the solution.

$$\begin{aligned}3sx_1 + 5x_2 &= 3 \\ 12x_1 + 5sx_2 &= 2\end{aligned}$$

Cramer's Rule and Inverses

To find A^{-1} , we typically row reduce the matrix $[A \ I]$

to get

$$[I \ A^{-1}]$$

Thus, the j th column of A^{-1} is a vector $\mathbf{x}$ satisfying

$$A\mathbf{x} = \mathbf{e}_j$$

where $\mathbf{e}_j$ is the j th column of I .

The i th entry of $\mathbf{x}$ is the (i, j) entry of A^{-1} .

Cramer's Rule and Inverses

$$A\mathbf{x} = \mathbf{e}_j$$

Thus, the (i, j) entry of A^{-1} is

$$x_i = \frac{\det A_i(\mathbf{e}_j)}{\det A}$$

If we evaluate $\det A_i(\mathbf{e}_j)$ by a cofactor expansion down column i , we see that

$$\det A_i(\mathbf{e}_j) = (-1)^{i+j} \det A_{ji} = C_{ji}$$

Cramer's Rule and Inverses

Thus, the (i, j) entry of A^{-1} is

$$\frac{C_{ji}}{\det A}$$

and we can express A^{-1} as

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

Cramer's Rule and Inverses

The matrix of cofactors

$$\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

is called the adjugate (or classical adjoint) of A .

Thus,

$$A^{-1} = \frac{1}{\det A} \text{adj} A$$

Example 4

Compute the adjugate of the matrix and use it to find the inverse.

$$A = \begin{bmatrix} 3 & 5 & 4 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

WARNING

Questions about the adjugate (classical adjoint) have been known to pop up on the FE exam!

If you remember that

$$A^{-1} = \frac{1}{\det A} \text{adj} A$$

then you can calculate the adjugate as

$$\text{adj} A = (\det A) A^{-1}$$

where A^{-1} was calculated by any method.

Determinants and Volume

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$.

If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.

Example 5

Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1,3,0)$, $(-2,0,2)$, and $(-1,3,-1)$.
